

Tend a Board Test Jig SOP

One episode tests six boards. It runs from six untested boards lying in a numbered rack, each with its own serial card, an empty test jig with its lid open, two empty bins and an empty log tray, to six boards tested and sorted, each one lying in the PASS bin or the FAIL bin, with its serial card laid in the matching half of the log tray, the rack empty and the jig empty with its lid open.

One board is finished before the next one is started. The order for each board never changes: **seat the board on the jig pins, clamp the lid, press TEST, read the pass or fail lamp, sort the board to its bin, log the serial.** No lid is clamped until the board is flat down on both pins, TEST is not pressed until the handle is latched, the lid is not opened until a lamp has lit, and no new board is touched until the last board is in its bin and its serial card is in the log tray.

One arm does all of it. The same gripper takes every board, seats it, works the lid and the handle, presses the button, lifts the board out, lays it in its bin and moves its serial card. Nothing in this task has to be held, because nothing moves on its own: the rack, the jig, the bins and the log tray all grip the table, the lid stays where it is put on its hinge, and a board dropped over both pins lies flat on the bed on its own. Only one arm works in this cell. Do not bring a second arm into the task.

The board is only ever held by its two bare edges, the short edges with no parts on them. The gripper never closes on the face of the board, never touches a part, a connector or a pin on it, and never lays a board face down anywhere but on the jig bed or in a bin. A part knocked off a board cannot be put back on camera, so every rule in this SOP is there to keep the gripper off the faces.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one object is held at a time, no grasp is changed while an object is off the table, and no board is ever carried over the open jig, over a bin it is not going into, or over a board already lying in a bin. Every board travels straight forward from its slot to the jig, and forward again from the jig to its bin.

The lamp does the deciding, not the arm. A board goes in the bin the lamp calls for, and in no other bin. A board is tested once. A board that fails is not seated again, not tested again and not moved to the PASS bin: it goes to the FAIL bin, its card goes to the FAIL half of the log tray, and the episode carries on with the next board. Nothing on a board is pushed, reseated, wiped or fixed on camera.

Each tool goes back to its own spot before the next step starts. The counts are fixed: six boards, one jig check, one seating, one clamp, one TEST press, one lamp read, one bin and one card per board. There is no second press, no reseating a board that has been tested and no moving a board from one bin to the other. A board or a card that lands off the table is left

exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the rack with all six board slots and all six card slots, the test jig with its lid, its handle, its TEST button and both lamps, both bins, the log tray, and the check board rest.
3. The jig stays in frame the whole time, close enough that a reviewer can see whether the board is down flat on both pins, whether the lid is shut, whether the handle is latched down, and which lamp is lit.
4. Both lamps stay in frame and unblocked the whole time, so a reviewer can see which one lights and when. The arm does not stand between the camera and the lamps while a test is running.
5. All six rack slots and all six card slots stay in frame and their numbers face the camera, so a reviewer can see which slot each board and each card came out of.
6. Both bins and both halves of the log tray stay in frame, with their labels facing the camera, so a reviewer can see which bin a board goes in and which half a card goes in.
7. The light falls across the jig and the bins without glare on the lamps or on the board faces, so a lit lamp and a serial label can both be read from above.
8. The arm is at home with its gripper open.
9. The tabletop is dry and clear of anything but the items in the Materials checklist.
10. The arm reaches every spot on the table, all six board slots, all six card slots, the jig bed, the lid handle, the TEST button, both bins, both halves of the log tray and the check board rest, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
11. The arm comes down straight from above onto every rack slot, onto the jig bed and into both bins, so a board is lifted and lowered without being tilted or swung in.
12. The arm reaches the lid handle from the front and can push it down and lift it up without the gripper touching the board on the bed.

13. The paths from the rack forward to the jig, and from the jig forward to each bin, are unobstructed. Carry one board along each path by hand during setup: it must clear the rack edge, the open lid, the lamps, the log tray and both bin rims.

Materials checklist

1. One rack lies flat at the back of the table, straddling the midline, with six board slots in one line, numbered 1 to 6 left to right, and one card slot in front of each board slot carrying the same number. The numbers face the camera.
2. The rack grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or put it in a holder. Nothing pins the rack during the episode.
3. Six boards, and no more, one lying flat in each board slot, parts up, with the serial label readable from above. Every board is the same part in the same size, so any board fits the jig.
4. Each board slot is wide enough for a board to lift straight up out of it without an edge rubbing the slot wall, and shallow enough that both bare edges of the board stand clear for the gripper to close on.
5. Every board carries two locating holes, one near each of two opposite corners, matching the two pins on the jig bed, so the board only drops on the pins one way round.
6. Every board carries a serial label on its top face, printed large enough to read on camera from above.
7. Every board starts undamaged: no part hanging loose, no bent pin on a connector and no crack in the board. Look at each board during setup and replace one that is damaged.
8. Six serial cards, one lying face up in each card slot, each printed with the same serial as the board in the slot behind it. Set them out during setup and read each pair: a card whose serial does not match its board is replaced before the episode starts.
9. One test jig sits flat at the middle of the table, straddling the midline, with a flat bed, two locating pins standing up out of the bed, a lid hinged along its back edge, a clamp handle on the front of the lid, a TEST button on the jig frame clear of the lid, and two lamps side by side on the frame above the bed, the green one marked PASS and the red one marked FAIL.
10. The jig grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or bolt it down. Nothing pins the jig during the episode.
11. The jig bed is empty at the start of the episode, its lid is open and standing back on its hinge, and both pins are straight, clean and clear of anything left on them.
12. The lid stays where it is put. Open it by hand during setup and let go: a lid that falls shut, or that will not stay back off the bed, is fixed or replaced.
13. The clamp handle latches down under one straight push and lets go under one straight lift. Work it by hand during setup with a board on the bed: a handle that has to be forced, worked twice, or held down to stay latched is fixed or replaced.

14. The TEST button lights one lamp, and one only, under one press. Both lamps stay off while the test is running and one lights when it is done.
15. Both lamps work. Check them during setup: the green PASS lamp and the red FAIL lamp each light bright enough to read on camera, and neither is stuck on or stuck off.
16. Two bins sit at the front of the table, clear of the jig, the left one marked FAIL and the right one marked PASS, both empty at the start of the episode. Their labels face the camera.
17. Both bins grip the table and are shallow enough that the gripper lays a board flat inside one without letting go above the rim.
18. One log tray sits flat at the front centre, between the two bins, marked in two halves, FAIL on the left and PASS on the right. Both halves are empty at the start of the episode.
19. One check board lies on its own rest at the front left, clear of the jig. It is marked so it cannot be mistaken for one of the six, it is known good and lights the green PASS lamp every time, and it is never laid in a bin.
20. The tabletop, the rack, the jig, the bins and the check board rest are all anti-static, and the arm is bonded to the same mat. A cell that is not bonded is not usable for this task.

Workspace layout

Three rows: the boards at the back, the jig in the middle, and the bins and the log tray at the front. Every board travels straight forward out of its slot onto the jig, and straight forward again from the jig into its bin.

Back row:

1. Rack spot, back centre on the midline, the rack lying flat with six board slots in one line numbered 1 to 6 left to right, each board lying flat and parts up, and one card slot in front of each board slot holding that board's serial card face up.

Middle row, left to right:

2. Check board rest, at the left, clear of the jig's left edge.
3. Jig spot, centre, straddling the midline, the test jig with its bed and two pins, its lid standing open on its hinge, its clamp handle to the front, its TEST button on the frame clear of the lid, and the green PASS lamp and red FAIL lamp on the frame above the bed.

Front row, left to right:

4. FAIL bin, front left, empty and with its label facing the camera.
5. Log tray, front centre, lying flat with its FAIL half on the left and its PASS half on the right, both empty.
6. PASS bin, front right, empty and with its label facing the camera.

A clear gap of at least a gripper's width sits between the rack and the jig, between the jig and each bin, and between the log tray and each bin, so a board lifts straight up out of its slot,

comes down onto the jig bed, and comes down into its bin without the gripper touching the rack, the lid, the lamps, the log tray or a board already in a bin. No carry passes over the open jig once the lid is up, over the lamps, or over a bin the board is not going into.

Vocabulary

- **Arm assignment:** one gripper does everything: every board, every clamp, every press, every card. This is a single arm task and no second arm takes any part in it.
- **Tool home:** the spot the check board returns to and lies flat on before the next step starts. The check board is never left on the jig bed, in a bin, in a rack slot or in the gripper between steps.
- **Release point:** the gripper opens over the jig bed when a board is down flat on both pins, inside a bin when a board is lying flat on the bottom, over the log tray when a card is lying flat in its half, or over the check board rest. Never over the rack, never over the lamps and never over open tabletop.
- **Stable placement:** the object stands still after release, square on its surface, and does not rock, tip, slide or lean on anything.
- **Board:** one of the six circuit boards that are tested. It is a flat rectangle with parts on its top face, two locating holes near opposite corners, and a serial label on the top face.
- **Bare edges:** the two short edges of the board, the strips with no parts on them. They are the grasp: the gripper closes on these and on nothing else.
- **Serial label:** the printed label on the board's top face that names the board. It faces up in the rack, on the jig and in the bin, so it can be read on camera at every point.
- **Serial card:** the card in the rack's card slot printed with the same serial as the board in the slot behind it. Moving it into the log tray is how the board's serial is logged.
- **Locating pin:** one of the two pins standing up out of the jig bed. The board's two holes drop over them, and the pins are never bent, pushed sideways or gripped.
- **Board down flat:** the board lying flat on the jig bed with both pins through their holes, no gap under it anywhere round its edge, and no corner standing up.
- **Lid:** the hinged cover over the jig bed. It stays back on its hinge when it is open and sits down flat on the jig when it is shut. It is never lowered onto a board that is not down flat.
- **Clamp handle:** the handle on the front of the lid. It is latched when it has been pushed straight down flat and stays down on its own. It is free when it has been lifted straight up and the lid can be raised.
- **TEST button:** the one button on the jig frame that starts the test. It is pressed once per board, and only with the lid shut and the handle latched.
- **Test running:** the time between the press and a lamp lighting. Both lamps are off. The gripper stays clear of the jig, the lid and the lamps until a lamp lights.

- **PASS lamp:** the green lamp. When it is the lamp that lights, the board goes in the PASS bin and its card goes in the PASS half of the log tray.
- **FAIL lamp:** the red lamp. When it is the lamp that lights, the board goes in the FAIL bin and its card goes in the FAIL half of the log tray.
- **Check board:** the marked known good board, seated and tested once at the start to prove the jig, the clamp, the button and the lamps. It always lights the green PASS lamp, and it is never laid in a bin and never logged.
- **Quarter turn:** the most any object is tilted or turned in the gripper. Taking a board, carrying it and laying it down all stay inside it. Nothing is ever turned over in the air.

Steps

Step 1: Check the jig with the check board

Goal: the check board seated once, clamped, tested, seen to light the green PASS lamp, and back on its rest with the jig bed empty and the lid open.

The check happens once, before the first board. If the clamp does not latch, the button does not fire or a lamp is stuck, every result in the episode is wrong, so the jig is proved on the check board first.

1.1 Take the check board

- **Check:** the jig bed is empty, the lid is open and both lamps are off.
- With the **gripper**, close on the two bare edges of the check board square, clear of the parts on its face, and lift it straight up off its rest until it is clear of the table.
- Keep it level, parts up, the whole way.

Check: the check board hangs level in the gripper and does not slip, and the gripper is on the bare edges only. If it hangs tilted, lay it back on its rest and take it again. Make no more than two corrective takes.

1.2 Run it through the jig

- Carry it over the jig bed, line its two holes up over the two pins, and lower it straight down until it lies flat on the bed. Open the gripper and lift clear.
- Close the lid down onto the jig, then push the clamp handle straight down until it latches and stays down on its own. Lift the gripper clear of the lid.
- Press the TEST button once, straight down, then lift the gripper clear of the jig and wait with the arm clear of the lamps until a lamp lights.

Check: the green PASS lamp is lit and the red FAIL lamp is off. If the red lamp lights, if no lamp lights, or if the handle did not latch under one push, end the episode and fix the jig during reset. The check board is not tested again.

1.3 Return the check board to its rest

- Lift the clamp handle straight up until it is free, raise the lid back on its hinge until it stays open on its own, and lift the gripper clear.
- With the **gripper**, close on the check board's two bare edges square, lift it straight up off both pins without tilting it, carry it back to its rest, lower it until it lies flat, release, and lift clear.

Check: the check board lies flat on its rest, parts up, the jig bed is empty, the lid is open, and the jig has not moved. One jig check, and one only. It is not run again between boards.

Step 2: Take one board from the rack

Goal: one board hanging level in the gripper by its two bare edges, parts up, clear of the rack.

Boards are worked in slot order, slot 1 first and slot 6 last. The rack grips the table, so nothing holds it while the board comes out.

2.1 Close on the two bare edges

- With the **gripper**, come down straight from above on the board in the next slot in order, and close on its two bare short edges, square to the board.
- The gripper closes on the edges and on nothing else. It never closes on the face of the board, never comes down on a part or a connector, and never pinches the board across its parts.

Check: the gripper is square on both bare edges and clear of every part. If it has landed on the face, open, lift clear and come down again. Make no more than two corrective takes.

2.2 Lift it straight out

- Lift the board straight up out of the slot until it is clear of the rack, before any sideways move.
- Keep it level, parts up, serial label readable. Do not tilt it, do not drag it up the slot wall, and do not slide it across the rack.

Check: the board hangs level in the gripper, parts up, and does not slip, nothing has come off it, and the rack has not moved. Leave the board's serial card where it is: the card is not moved until the board is in its bin.

Step 3: Seat the board on the jig pins

Goal: the board lying down flat on the jig bed with both pins through their holes.

The board is lined up in the air first and only then lowered, so no hole has to be dragged sideways onto a pin.

3.1 Line the holes up over the pins

- Carry the board forward to the jig and hold it level above the bed, a board's height clear of it.

- Turn it no more than a quarter turn until its two holes stand over the two pins. The board only fits one way round, so if the holes will not line up, turn it the quarter turn that brings them over the pins.

Check: both holes stand over both pins and the board hangs level and square above the bed. If it would take more than a quarter turn, lay it back in its slot, take it again, and turn it once. Make no more than one correction.

3.2 Lower it onto the pins

- Lower the board straight down until both pins are through their holes and the board lies flat on the bed under its own weight. Stop there.
- Both pins go in together. Do not put one hole on its pin first and lean the other one down after it, and do not slide the board across the bed to find the pins.
- Open the gripper and lift straight clear of the board.

Check: the board lies down flat on the bed, both pins through their holes, no corner standing up and no gap under it, and it does not rock when the gripper lets go. If it is not flat, lift it straight up clear of both pins and lower it again. Make no more than one correction.

Step 4: Clamp the lid

Goal: the lid shut on the jig and the clamp handle latched down on its own.

The lid is only ever closed on a board that is already down flat. A board that is standing on a pin, or lying proud of the bed, is crushed by the lid.

4.1 Close the lid

- **Check:** the board is down flat on both pins and the gripper is clear of the bed.
- Bring the lid down from its hinge onto the jig in one steady move until it sits down flat, then lift the gripper clear of it.
- Do not drop the lid, do not slam it, and do not close it part way and press it down.

Check: the lid sits down flat all the way round with no gap at the front, and the board has not shifted or lifted under it. If the lid will not sit flat, raise it, check the board is down flat, and close it again. Make no more than one correction.

4.2 Latch the handle

- Bring the gripper down square on the clamp handle and push it straight down in one steady move until it latches and stays down on its own.
- Lift the gripper straight off the handle and clear of the lid.
- One push, straight down. Do not work the handle up and down, do not push it sideways, and do not keep hold of it to hold it down.

Check: the handle is latched flat down and stays down with nothing touching it. If it springs back up, the jig is not clamped: end the episode and fix the handle during reset. Do not press TEST on an unlatched jig.

Step 5: Press TEST and read the lamp

Goal: one press of TEST, and one lamp lit, either the green PASS lamp or the red FAIL lamp.

The lamp is the result. It is read once, with the arm clear of the jig, and it is what decides the bin. Nothing else decides the bin.

5.1 Press TEST once

- **Check:** the lid is shut and the handle is latched down on its own.
- Bring the gripper down square on the TEST button, press it straight down once, and lift straight off.
- One press only. Do not hold the button down, do not press it twice, and do not press it again while the test is running.

Check: the button has been pressed once and both lamps are off, so the test is running.

5.2 Wait and read the lamp

- Move the arm clear of the jig, the lid and both lamps, and hold it there while the test runs.
- Wait, without touching anything, until one lamp lights.
- Read the lamp: green PASS or red FAIL. Do not lift the handle, open the lid or touch the board until a lamp is lit.

Check: exactly one lamp is lit and it is in clear view of the camera. If both lamps light, or no lamp lights, end the episode and fix the jig during reset. The board is not tested again either way.

Step 6: Sort the board to its bin

Goal: the board lying flat in the bin the lamp called for, parts up and serial label readable.

Green goes right to the PASS bin, red goes left to the FAIL bin. The bin is chosen by the lamp that lit and by nothing else.

6.1 Unclamp and open the lid

- Bring the gripper under the clamp handle and lift it straight up until it is free of its latch.
- Raise the lid back on its hinge until it stays open on its own, and lift the gripper clear of it.

Check: the handle is free, the lid stands open on its own, and the board is still lying flat on both pins. The lamp stays lit and in view.

6.2 Lift the board off the pins

- With the **gripper**, close on the board's two bare edges square, clear of the parts, and lift it straight up off both pins until it is clear of them, before any sideways move.
- Keep it level, parts up. Do not tilt it and do not drag it across the bed.

Check: the board hangs level in the gripper, both pins are clear and straight, and the jig bed is empty.

6.3 Lay it in the bin the lamp called for

- Carry the board forward to the PASS bin on the right if the green lamp lit, or to the FAIL bin on the left if the red lamp lit.
- Lower it straight down inside the bin until it lies flat, parts up and serial label readable, then open the gripper and lift clear.
- Lay it flat beside the boards already in the bin, not on top of one, and never let go above the rim.

Check: the board lies flat in the bin the lamp called for, parts up, and it does not rock or lean on another board. A board in a bin is never moved to the other bin and never taken back to the jig, whatever the check finds. If a board goes in the wrong bin, end the episode and let review flag it.

Step 7: Log the serial

Goal: that board's serial card lying face up in the half of the log tray that matches the lamp.

The card is moved last, after the board is in its bin, so the card in the tray always stands for a board that is already sorted.

7.1 Take the serial card

- Come back empty to the rack and, with the **gripper**, close on the serial card in the card slot with the same number as the board's slot.
- Lift it straight up out of the slot, keeping it flat and face up.

Check: the card in the gripper carries the same serial as the board just binned, and it came out of the card slot with the same number as that board's slot. Only one card is lifted.

7.2 Lay it in the matching half

- Carry the card forward to the log tray and lower it into the PASS half on the right if the green lamp lit, or into the FAIL half on the left if the red lamp lit.
- Lower it until it lies flat and face up with its serial readable, then open the gripper and lift clear.
- Lay it flat beside the cards already in that half, not on top of one.

Check: the card lies flat and face up in the half that matches the lamp, its serial can be read, and the card slot it came out of is empty. A card in the tray is never moved to the other half.

Step 8: Do the next board

- Come back empty to the rack and take the board in the next slot in order.
- Run Steps 2 to 7 again for that board.
- Six boards, worked one at a time from slot 1 to slot 6. A board is in its bin and its card is in the log tray before the next board is touched.
- The board slots stay in step with the card slots: the board from slot 1 is logged with the card from card slot 1, and the board from slot 6 with the card from card slot 6.
- The check board is not tested again. The jig is proved once per episode.

Check: after each board, the rack has one fewer board and one fewer card in it, one more board lies in a bin, one more card lies in the log tray, the jig bed is empty and the lid is open. No board is tested twice or skipped.

Step 9: End the episode

- Confirm the rack: all six board slots and all six card slots empty.
- Confirm the bins: six boards in all, each one lying flat and parts up in the bin its lamp called for.
- Confirm the log tray: six cards in all, each one lying flat and face up in the half that matches its board's bin, and the counts match, the PASS half holding as many cards as the PASS bin holds boards.
- Confirm the jig: bed empty, both pins clear and straight, lid open, handle free.
- Confirm the lamps: both off, with nothing left standing in front of them.
- Confirm the check board: lying flat on its rest, parts up.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the arm home with its gripper open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the boards out of both bins by hand, one at a time, keeping each one level and parts up.
2. Look at every board. Set aside any board with a part knocked loose, a bent connector pin or a crack: it is not fixed and it is not used again.
3. Lay six boards back in the rack, one to a slot, flat and parts up with the serial label readable, and replace any board that was set aside so the rack holds exactly six.

4. Take the cards out of both halves of the log tray and lay them back in the card slots, face up, each card in the slot in front of the board with the same serial. Read every pair before the next episode.
5. Pick up any board or card that fell off the table. Bin a card that has been on the floor and print a new one. Check a board that has been on the floor and set it aside if anything on it has moved.
6. Look at the jig bed: clear of anything left on it, both pins straight, clean and unbent. Replace a bent or loose pin.
7. Work the lid and the clamp handle by hand with a board on the bed: the lid must sit down flat, stay back on its hinge when it is opened, and the handle must latch under one straight push and free under one straight lift.
8. Seat the check board by hand, clamp it, press TEST once and confirm the green PASS lamp lights and the red FAIL lamp stays off. Unclamp, lift the check board out and lay it flat back on its rest.
9. Confirm both lamps still light bright enough to read on camera and neither is stuck on or stuck off.
10. Confirm the rack, the jig, both bins, the log tray and the check board rest all still grip the table, and that the slot numbers, the bin labels and the log tray labels still face the camera.
11. Wipe the tabletop, the rack, the jig bed and both bins clean and dry, and confirm the mat and the arm are still bonded.
12. Confirm both bins and both halves of the log tray are empty and the light still falls across the jig without glare on the lamps.
13. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** a board is tested before the check board run, the lid is closed before the board is down flat on both pins, TEST is pressed before the handle is latched, the lid is opened before a lamp lights, a serial card is moved before its board is in a bin, a new board is taken out of the rack while the last one is still on the jig, or a step is skipped.
- **SOP rule broken:** Steps 1 to 8, prove the jig once, then for each board seat it, clamp it, press TEST, read the lamp, bin it, log it, in that order.
- **Coaching note:** check board first, then one board all the way to its bin and its card before the next one is touched. Seat it, clamp it, then press.

Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the rack, the jig, the lid or a board, works the handle, or presses the button. Any object taken, carried, clamped or pressed by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 9, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.
- **Coaching note:** one arm only. If a second arm has worked, the episode is a bimanual episode wearing the wrong label.

Violation: Object held or steadied

- **Visible cue:** the gripper presses on the rack, the jig, a bin or the log tray to keep it still while working, holds the lid down while the handle is pushed, keeps hold of the clamp handle to hold it latched, or holds the board on the bed while the lid comes down, instead of letting the table and the jig do it.
- **SOP rule broken:** Steps 1 to 6, nothing in this task is held: every fixture grips the table, the lid stays where it is put, the handle stays latched on its own and a board over both pins lies flat on its own.
- **Coaching note:** if something has to be held down, it was not set up right. Let go and let the fixture do its job.

Violation: Board picked up wrong

- **Visible cue:** the gripper closes on the face of a board, comes down on a part or a connector, pinches the board across its parts, takes two boards at once, takes a board out of slot order, drags a board up the slot wall or slides it across the rack, or lifts a board sideways before it is clear of the slot or clear of both pins.
- **SOP rule broken:** Steps 2.1, 2.2 and 6.2, close square on the two bare edges only, and lift the board straight up clear before any sideways move, one board at a time and in slot order.

- **Coaching note:** the two bare edges and nothing else, then straight up. Everything that touches a face costs a board.

Violation: Board seated wrong on the pins

- **Visible cue:** the board is lowered with a hole off its pin, one hole is put on its pin and the other leaned down after it, the board is slid across the bed to find the pins, the board is left standing up on a pin or proud of the bed, or it is dropped onto the bed from height.
- **SOP rule broken:** Steps 3.1 and 3.2, line both holes up over both pins in the air and lower the board straight down until it lies flat on the bed.
- **Coaching note:** line it up before it touches. A board that has to be dragged sideways onto a pin is not seated.

Violation: Lid or clamp worked wrong

- **Visible cue:** the lid is closed on a board that is not down flat, dropped or slammed shut, closed part way and pressed down, or raised so far it comes off its hinge; or the handle is pushed sideways, worked up and down, forced when it will not latch, or left unlatched when TEST is pressed.
- **SOP rule broken:** Steps 4.1 and 4.2, close the lid in one steady move onto a board that is down flat, then push the handle straight down once until it latches on its own.
- **Coaching note:** flat board, then lid, then one straight push on the handle. A lid closed on a proud board breaks the board and the jig.

Violation: TEST pressed wrong, or pressed more than once

- **Visible cue:** TEST is pressed with the lid open or the handle unlatched, pressed and held down, pressed twice for one board, pressed again while the test is running, or pressed again after a lamp has lit.
- **SOP rule broken:** Step 5.1, one press, straight down, once per board, and only with the lid shut and the handle latched.
- **Coaching note:** one press per board. A board is tested once and the first lamp is the answer.

Violation: Jig disturbed while the test is running

- **Visible cue:** the gripper touches the jig, the lid, the handle, the board or the button while both lamps are off, the lid is opened or the handle lifted before a lamp lights, or the arm stands in front of a lamp so the result cannot be seen.
- **SOP rule broken:** Step 5.2, move clear of the jig and both lamps after the press and wait, touching nothing, until one lamp lights.
- **Coaching note:** hands off while it runs. The wait is part of the step, not a gap in it.

Violation: Board sorted to the wrong bin

- **Visible cue:** a board goes in the PASS bin when the red lamp lit, goes in the FAIL bin when the green lamp lit, goes in a bin with no lamp lit, is moved from one bin to the other, or is laid on top of a board already in a bin instead of flat beside it.
- **SOP rule broken:** Step 6.3, the lamp that lit chooses the bin: green to the PASS bin on the right, red to the FAIL bin on the left, laid flat and parts up.
- **Coaching note:** the lamp decides, not the arm. Once a board is in a bin it stays in that bin.

Violation: Serial logged wrong

- **Visible cue:** a card is laid in the half that does not match the lamp, a card is moved before its board is in a bin, the card from the wrong slot is taken, two cards are taken at once, a card is laid face down or on top of another card, a card is moved from one half to the other, or a board is binned and its card left in the rack.
- **SOP rule broken:** Steps 7.1 and 7.2, take the card from the card slot with the board's number after the board is binned, and lay it flat and face up in the half that matches the lamp.
- **Coaching note:** one board, one card, same number, same result. A card in the wrong half is a wrong record.

Violation: A tested board worked on again

- **Visible cue:** a board that has been tested is put back on the jig, tested a second time, reseated on the pins, or has a part pushed back, a pin straightened or the board wiped or blown on camera.
- **SOP rule broken:** Steps 5 and 6, a board is tested once and a board that is off the jig is never seated again on camera.
- **Coaching note:** once it is tested, it is sorted. Rework belongs in the reset, not in the episode.

Violation: A board carried over the wrong thing

- **Visible cue:** a board passes over the open jig once the lid is up, over the lamps, over a bin it is not going into, over a board already lying in a bin, over the log tray, or back over the rack after it has been lifted out.
- **SOP rule broken:** Steps 2, 3 and 6, each board travels straight forward from its slot to the jig and forward again to its own bin, and over nothing else.
- **Coaching note:** route around, not across. A board dropped on the lamps ends the episode for every board after it.

Violation: Board or card laid down off its spot

- **Visible cue:** a board is set down on the jig frame, on the tabletop, on a bin rim, on the log tray or on the rack rim, or a card is set down anywhere but its card slot or its half of the log tray.
- **SOP rule broken:** Steps 2 to 7, a board lies flat only in its rack slot, on the jig bed or in a bin, and a card lies flat only in its card slot or in the log tray.
- **Coaching note:** there are only three places a board belongs and two places a card belongs. Bare table is not one of them.

Violation: Bad pick, carry, or release

- **Visible cue:** an object is picked up before the last one is set down, two objects are held at once, the gripper shifts or seats its hold again while the board is off the table, a board slides along the table or the bed with no daylight under it, a board hangs tilted because the grip was taken off square, an object is released above a bin rim or above the log tray, or an object is turned past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 7, one object at a time, lifted straight up clear of the surface before any horizontal motion, gripped square on its bare edges, turned no more than a quarter turn, and released only when it is lying flat.
- **Coaching note:** set it down to change the grip, and lift before the carry. A board that hangs tilted was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a board, a card, the rack, the jig, a bin or the log tray is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 7, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** the gripper opens while the check board is still above its rest, the check board is left on the jig bed, in a bin, in a rack slot or on the tabletop, or it is still in the gripper when the next step begins.
- **SOP rule broken:** Step 1.3, the check board returns to its rest and lies flat there before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower it until it lies flat, then release.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, the gripper or a carried board strikes the rack, the jig, the lid, a lamp, a pin, a bin or a board already in a bin and knocks, tips, slides or displaces it; a board or a card falls off the table or out of the gripper; or a board lands short of the jig bed or of its bin, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 7, keep each board and card on a clear travel path and release only when the object is lying flat on its spot.
- **Coaching note:** watch the whole path, not just the target. The lamps and the seated boards are the things worth steering widest of.

Violation: Manipulation outside the camera frame

- **Visible cue:** a board is taken, seated, clamped, lifted or binned partly or wholly outside the environment camera frame, a card is moved outside the frame, or the arm stands between the camera and the jig while the seating, the handle or a lamp is what has to be seen.
- **SOP rule broken:** Steps 1 to 7, every manipulation happens inside the camera frame, and the jig, the lamps and the board being worked stay visible.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and keep the arm clear of the jig.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a board still in the rack, a card still in a card slot, a board still on the jig bed, the lid still shut or the handle still latched, a lamp still lit, a board in the wrong bin, a card in the wrong half, the check board off its rest, the arm short of home, or the gripper not fully open.
- **SOP rule broken:** Step 9, confirm the rack, the bins, the log tray, the jig, the lamps and the check board, then return the arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.

- Defective board at episode start: one that already carries a loose part, a bent connector pin or a crack, one whose serial label cannot be read, or one whose holes do not drop over the jig pins.
- Defective jig: a bent or loose locating pin, a bed that will not let a board lie flat, a lid that falls shut or will not stay back on its hinge, or a clamp handle that has to be forced or will not stay latched.
- Defective button or lamp: a TEST button that does not fire under one correct press, a lamp that is stuck on or stuck off, both lamps lighting at once, or no lamp lighting at all.
- Test never finishes: both lamps still off after a correct press, so there is no result to sort on.
- Defective rack: one that slides, turns or lifts under a correct pick, a board slot so tight that a board cannot lift straight out, or a slot that holds a board below the reach of the gripper.
- Defective bin or log tray: one that slides, turns or lifts under a correct place, or one so deep that a board or a card cannot be laid flat inside it.
- Defective check board: one that no longer lights the green PASS lamp in a jig that is otherwise good.
- Static event: a mat, rack, jig or arm that has come unbonded, or a board handled outside the anti-static setup. Discard the board as well as the episode.
- Glare, shadow or a light flickering over the jig that stops a lamp, the handle or the seating being seen. Fix the lighting before the next episode.
- Rack holding fewer or more than six boards or six cards at episode start, a card whose serial does not match the board in front of it, a bin or a log tray half that is not empty, or a board left on the jig bed. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take the check board from its rest
2. Run the check board through the jig
3. Return the check board to its rest
4. Take one board from its rack slot
5. Seat one board on the jig pins
6. Close the lid and latch the clamp handle
7. Press TEST once
8. Wait for and read the pass or fail lamp
9. Unclamp and open the lid
10. Lift one board off the pins
11. Lay one board in the PASS bin or the FAIL bin
12. Lay one serial card in the matching half of the log tray

13. Return the arm home and end the episode